

Killer-Day Mitigation — S115 Definitive Research Note

Date: 2026-09-10 (S115, chat B) · **Branch state:** files only, not committed tonight per instruction **Data:** data/options_sim/backtest_full/killer_context.csv (1,080 days, 2022-05-16 → 2026-09-04), rows.csv (vix252 streams), fly_rows.csv **Scripts:** scripts/killerday/ (forensics per day, 6 family sweeps, stack, adversarial audit) · **Results CSVs:** data/options_sim/backtest_full/killerday/ **Split everywhere below:** train = 2022–2024 (661 days), test = 2025–2026 (419 days). Baselines (train/test): ic -21,720/-3,203 · fly -125,857/-74,691 · puts_band +11,781/+3,146.

1. Executive summary — can catastrophic days be mitigated?

Partially — and less than the raw sweeps claim.

- **33 killer days** were forensically dissected (news reconstruction + local trade rows). **22 of 33 (67%) were flaggable before the 08:30 CT entry** — by the vol regime, the calendar, or our own entry credits. **11 of 33 (33%) were pure intraday ambushes** with a quiet open, calm VIX, and no scheduled event: no day-level filter can touch them.
- **~26 of 33 killer days opened with |gap| < 0.5%.** The quiet open is the trap signature, not a safety signal.
- The rule sweeps produced dozens of "ROBUST" filters, but **adversarial verification (random-skip Monte-Carlo null + Bonferroni ×395 comparisons, scripts/killerday/adversarial_audit.py) kills almost all of them: only 21/174 audited rule×book combos survive, and 16 of those 21 are on the fly book.** Zero ic single rules and zero puts_band single rules survive as proven.
- **The one strongly confirmed mechanism** ($p_{\text{bonf}} \approx 2e-6$): the **fly book is destroyed in elevated/rising-vol regimes** (skip_vix>25 fly, prior_ret≤-1.0, cr_p>3.0, total_cr>3.0 all confirm the same factor). The correct action is **retire the fly book** (-\$200k baseline, still -\$20k after its best stack), not filter it.
- **The one safe-by-construction rule:** the **puts_band -1000 intraday flatten** — pessimistic fill bound ≈ \$0 cost, MID estimate +3,183/+3,966 train/test, 0 winners lost. Adoptable now (advisory-desk terms).
- **The best unproven hypotheses** — surgical **ic call-leg drop when call credit is rich (>2.0–3.0)** (+4.2–4.8k train / +6.4–7.7k test, joint p 0.003–0.005, Bonferroni-dead) and the **FOMC 12:30 CT flatten** (currently an accounting upper bound, ic +12.1k/+5.6k vs honest skip_FOMC +2.1k/+2.0k) — go to **stage-2 tomorrow:** minute-level ThetaData sim on the v2 engine.
- **Headline stack numbers** (ic -24.9k → +45.5k total) are **selection artifacts** (greedy accepted/pruned on marginal TEST delta) and must not be booked. Direction is real on a few members; magnitudes are not validated.
- Per the desk hard rule, everything here is **advisory-only — no trade is ever blocked**; survivors ship as flags and are judged on the forward record.

2. The killer-day dossier

2a. One row per day

Date	Cause (≤8 words)	Sched. event?	Flag by 08:30?	Signals that fired
2022-05-20	OpEx whipsaw at bear-market touch, no news	OpEx	YES	EM 1.85%, VIX 29, post -4% day, put cr 0.95 vs call 4.80
2022-06-07	Down-gap faded; one-way rip through calls	—	no	call cr 0.10 (thin gate), gap-down w/ flat VIX
2022-06-09	Hawkish ECB premarket; CPI-eve grind-down	ECB + claims	YES	ECB out pre-entry, put:call credit 8:1 (2.85/0.35), CPI-eve
2022-11-08	Election drift; FTX rescue reversal	Midterms (not in cal)	no	election not in calendar; FTX crypto-only premarket
2023-02-14	Hot CPI; two-way travel, flat close	CPI	YES	CPI flag + hot print pre-entry, put cr 5.4 vs 0.85, EM rich vs quiet tape
2023-03-13	SVB Monday; regional-bank whipsaw	— (bank crisis)	YES	VIX 24.8 rising, put cr 6.70 vs 0.95, gap-down despite backstop
2023-05-02	Regional-bank rout on FOMC-eve	FOMC-eve, JOLTS	YES	call cr 0.00 (broken gate), FRC seizure 24h old, FOMC-eve
2023-09-15	Triple witching + TSMC + UAW drift	Quarterly OpEx	YES	put:call credit 44:1 (2.20/0.05), VIX 12.8 complacency, OpEx
2023-10-12	Hot CPI + tailed 30y auction	CPI + 30y auction	YES	CPI flag, hot print pre-entry, 3y/10y auctions already tailed
2023-12-20	ODTE put-flow flush at 52wk highs	—	no	none; VIX 12.5, gap -0.08%, call cr 0.05 only
2024-04-09	CPI-eve de-risk round trip	— (CPI next day)	no	CPI-eve not flagged; gold record/Iran watch soft only
2024-04-30	Hot ECI on FOMC-eve, month-end	ECI, FOMC-eve	YES	ECI hot at 07:30 CT, call cr 0.05 vs put 1.75, month-end

Date	Cause (≤8 words)	Sched. event?	Flag by 08:30?	Signals that fired
2024-05-31	MSCI rebalance two-sided whipsaw	PCE, MSCI rebal	no	rebalance calendar only; VIX 14.5, thin credits
2024-08-01	ISM 46.8 ambush after claims spike	ISM (not in cal), claims	YES	claims 249k pre-entry, BOJ/yen stress, post-FOMC euphoria entry
2024-12-18	FOMC hawkish cut, dot-plot shock	FOMC + SEP	YES	FOMC flag; VIX creep 13.8→15.9 at highs; call cr 0.50 into binary
2024-12-20	Cool PCE relief rip, triple witching	PCE + OpEx	YES	PCE out pre-entry, VIX 24 falling post-shock, call cr 0.2 vs put 2.6
2025-02-21	OpEx + PMI/UMich stagflation prints	OpEx + data 08:45/09:00	no	data slate post-entry; call cr 0.05; VIX 15.7 calm
2025-02-28	Month-end melt-up through calls	PCE, Zelensky, mo-end	YES	prior range 2.30% vs EM 1.33%, VIX 21 +2, desk bleeding 4 days
2025-03-03	Tariff D-day eve; afternoon confirmation	ISM + tariff deadline	YES	tariff effective-date public since 2/27, realized >> implied
2025-03-26	Midday auto-tariff announcement	—	no	VIX 17 falling, gap -0.09%; call cr 0.10 only (wrong wing)
2025-04-07	Crash day 3; fake-pause 8% whipsaw	—	YES	VIX 45→60 premkt, Nikkei -7.8%, EM 2.9%, put cr 12.3 vs 0.4
2025-04-08	Relief gap fully reversed; 104% confirmed	— (de facto deadline)	YES	VIX 47, gap +2.6% into unresolved binary, call cr 13.9 vs 0.35
2025-04-09	Real 90-day pause; +9.5% melt-up	— (FOMC min., 10y auct.)	YES	VIX 52.3, EM 3.3%, back-to-back 7–8% ranges, fat 2-sided credits
2025-10-10	Trump China-tariff post from ATH	— (shutdown blackout)	no	VIX 16.4, gap +0.08%; only soft tell = unanswered MOFCOM escalation
2025-10-22	Reuters software-curb exclusive midday	—	no	call cr -0.05 (broken); gold -6% prior day (soft); VIX 17.9 falling
2025-12-12	AVGO -11%; Oracle/OpenAI delay report	—	no	VIX 14.9 falling day-after-record; call cr 0.00; single-name only
2026-01-21	Greenland walk-back relief rip	— (Trump at Davos)	YES	day-after -2.06% shock, VIX 20 +4.2, call cr 1.45 = richest in window
2026-02-12	AI-displacement contagion de-gross	— (claims benign)	no	credits 0.2/0.3 thin; software rout was sector-level, not in features
2026-03-09	Oil-shock morning; peace-headline V-reversal	—	YES	VIX 29.5 +24%, WTI→\$119 overnight, gap-down on -1.33% day, EM 1.86%
2026-03-31	Iran peace melt-up +2.9%	Cons. conf. (minor), Q-end	YES	VIX 30.6 war regime, gap +0.82% after -8%, put cr 0.05 vs call 3.90
2026-06-09	Midday AI air pocket, full recovery	—	YES	prior2_ret -2.64% (NFP crash), put cr 0.05/0.25, CPI-eve stacking
2026-06-17	Warsh debut FOMC; dot-plot hike flip	FOMC + SEP + debut	YES	FOMC flag; event credits 1.7/1.6 vs 0.45/0.5 prior day; VIX silent
2026-07-29	FOMC hold; 30y-yield-spike dump	FOMC	YES	FOMC flag; put cr 3.0 vs call 0.7; EM 85pts on flat tape

2b. Narratives by type

Type A — scheduled-event kills (the calendar knew). FOMC days are the purest case: 2024-12-18 (hawkish-cut dot plot, -2.95%, VIX +74%), 2026-06-17 (Warsh debut, dot plot flipped to a hike), 2026-07-29 (hold read as behind-the-curve, 30y to 2007 highs). In all three the morning was dead flat and 100% of the damage came after the 13:00 CT statement — the 08:30 WAIT gate structurally cannot touch it. CPI killed twice (2023-02-14 two-way travel with a flat close — all fly, every IC settled green; 2023-10-12 where the real killer was the tailed 30-year auction at 12:00 CT). Data our calendar does not carry did the rest: ECB (2022-06-09), ISM (2024-08-01, 2025-03-03), ECI (2024-04-30), OpEx/rebalance days (2022-05-20 ~\$1.9T, 2023-09-15, 2024-05-31 MSCI, 2024-12-20 ~\$6.6T, 2025-02-21), elections (2022-11-08 — event column blank).

Type B — crash/headline-regime continuation (the tape knew). The April 2025 tariff cluster (04-07 fake-pause 8% whipsaw, 04-08 relief gap fully reversed, 04-09 real pause +9.52%), the 2026 Iran war (03-09 oil to \$119 then V-reversal, 03-31 peace melt-up +2.91%), SVB Monday (2023-03-13), the Greenland walk-back (2026-01-21), day-2-after-crash (2026-06-09), and the month-end squeeze inside a deteriorating tape (2025-02-28). Common signature: VIX ≥ 20–52 and/or prior ranges ≥ 2× EM, extreme one-sided credits (8:1 to 44:1), unresolved binary headline risk. Every mechanical vol/regime feature we carry fired. **Note the direction lesson: 8+ of these kills were UPSIDE (calls run over) — a downside-only defense loses a third of the time.**

Type C — intraday ambushes (nothing knew). 2025-10-10 (Trump post at 09:57 CT from 2 pts off ATH, VIX 16),

2023-12-20 (0DTE put-flow cascade, no news at all), 2026-02-12 (AI-displacement contagion), 2025-03-26 (midday auto-tariff presser), 2025-10-22 (Reuters exclusive), 2025-12-12, 2022-06-07, 2022-11-08, 2024-04-09, 2024-05-31, 2025-02-21. Signature: gap < 0.3%, VIX < 18 and often falling, no Tier-1 event, frequently a broken/near-zero credit on ONE side (the only in-system tell, and it flags the structure, not the day). These are unfixable at day level; they are the case for the intraday circuit breaker and for sizing as the primary tail control (Cboe/Schwartz, Vilkov 0DTE).

3. What the forensics say statistically

- **22/33 (67%) detectable by 08:30 CT**; 11/33 (33%) ambushes.
- **What flagged the detectable ones** (overlapping):
- **Vol regime** (VIX $\geq$ ~20–25, spike, or prior-range $\geq 2\times$ EM): ~13 days — the whole Type-B family.
- **Calendar carried by the engine** (FOMC/CPI/NFP): only **5 of 33** (3 FOMC + 2 CPI). **Calendar gaps caused misses on at least 8 more**: ECB, ISM ($\times 2$), ECI, OpEx/witching ($\times 5$), MSCI rebalance, elections, Treasury auctions, tariff effective-dates.
- **Own entry credits** (broken ≤ 0 , thin < 0.10, or extreme skew $\geq \sim 3:1$): fired on ~17 of 33 days — the single most frequent tell, and the only one that also fires on some ambush days (10-22, 12-12, 02-21, 03-26 flagged the structure even when the day looked calm).
- **The quiet-open trap**: ~26/33 killer days opened with $|\text{gap}| < 0.5\%$. Gap-based filters alone are near-useless on this population.
- **Direction**: ~8 kills were upside/call-side, ~10 were two-sided whipsaws, remainder downside. Several losses were **stop-manufactured, not settlement losses** (2024-12-20 close never breached the short call; 2026-06-09 every stopped put finished OTM; 2023-02-14 all ICs settled green; 2022-11-08 and 2024-05-31 EM-width ICs survived while flies died) — the loss mode is often travel + stop width vs credit, which no day-skip fixes.
- **Timing**: on FOMC days and several headline days, all damage is post-13:00 CT; on Type-B days it is continuous; on ambush days it is a 30–90-minute midday air pocket.

4. Rules that SURVIVED adversarial verification (per book)

Verification = re-run reproduction (all 7 scripts regenerate CSVs byte-identical, no lookahead found) **plus** random-skip Monte-Carlo null + Bonferroni across ~395 comparisons (adversarial_audit_20260910.csv). Survivors are few.

fly — one confirmed mechanism, terminal conclusion

Rule	Train Δ	Test Δ	p_bonf	Verdict		
skip_vix>25	+37,188	+23,616	2e-6	CONFIRMED — genuine selectivity (skipped days avg $-\$483$ vs $-\$186$ base)		
prior_ret ≤ -1.0 /\	prior_ret\	≥ 1.0	+30,853/+60,109	+29,832/+34,610	3e-5 / 0.023	CONFIRMED — fly needs quiet days
skip_day_if_cr_p>2.5/3.0	+35,011/+31,808	+35,896/+34,517	0.047/0.0002	CONFIRMED — fly dies when puts are bid		
skip_day_if_total_cr>3.0	+51,869	+55,530	0.001	CONFIRMED — vol-regime selectivity		
vixchg>3 / accel2d>3 / 2day_both_up1 / vix-vs-5dma/10dma	various	various	0.0002–0.012	CONFIRMED — same rising-vol factor, count as ONE		

Economic story: all of these are correlated expressions of a single factor — the ATM fly is structurally short realized vol and gets destroyed whenever vol is elevated or rising. The stacked fly still ends **−\$20.2k** total with 903 days zeroed. **Action: retire the fly book**. That is the entire actionable content of the fly column.

puts_band — one safe rule

Rule	Train Δ	Test Δ	Bounds	Verdict
flatten at day P&L ≤ -1000 (intraday breaker)	+3,183 (MID)	+3,966 (MID)	PESS $-47/0$ · OPT $+6,799/+7,685$	CONFIRMED by dominance — worst-case fill cost $\approx \$0$, 0 winners lost
skip_day_if_total_cr<1.0 (dead-tape floor)	+1,165	+5,145	joint p 0.024, fails $\times 395$	WEAKENED — coherent, cheap hygiene; dollar value unproven

Economic story: the credit band already filters entries, so a $-\$1,000$ day means both put spreads stopped on a real selloff with nothing left to recover — flattening forfeits ~nothing even under worst fills. This is the only rule in the entire study whose **sign is safe without further data**.

ic — nothing proven; two live hypotheses

Hypothesis	Train Δ	Test Δ	Status
ic_drop_call_leg_if_cr_c>2.0/2.5/3.0 (surgical)	+4.8k/+4.2k/+3.8k	+7.7k/+7.3k/+6.4k	WEAKENED — direction consistent (test>train all thresholds, ex-April +4.5–5.7k), joint p 0.0034–0.0047, but train contribution $\approx$ random and the 3 thresholds are one correlated rule. Stage-2 candidate.
flatten at day P&L ≤ -1000 (breaker)	+7,398 (MID)	+10,547 (MID)	WEAKENED — 40/60 killer days caught, 0 winners lost, but PESS bound is -16,435/-6,380 : the sign flips under pessimistic fills. Stage-2 decides.
FOMC close-12:30	+12,132 (bound)	+5,627 (bound)	Not tested — accounting upper bound; honest bracket is [skip_FOMC +2,080/+2,042, bound]. Stage-2 decides.

Zero ic whole-day skip rules survive. The S114 rich-credit premise holds asymmetrically at best: call-side only, as a leg-level hypothesis.

5. Rules that FAILED — and why (honesty section)

The sweep's ROBUST criterion ($\Delta > 0$ in both halves, no significance test, no multiplicity control) **approves noise by design**. Failure modes, with the worst offenders:

- **April-2025 cluster fits.** The tariff week (≤ 7 skips) carries 50–100% of test deltas across the vol/gap/range families. Ex-April: skip_vix>25 ic **-378**, total_cr>3.0 ic **-1,813**, vix/5dma>1.10 ic **-3,387**, prior_range ≥ 3 &gap **-2,771**, prior_range ≥ 4.0 combined **exactly 0**. These are one week wearing a rule costume.
- **Single-day deltas.** Event-family test deltas have top-1-day shares ≥ 1.0 : skip_FOMC ic 1.32 (ex-top1 negative), skip_ANY_EVENT ic 2.3, skip_NFP ~ 1.0 –1.2. One day exceeds the whole claimed OOS edge $\rightarrow$ REFUTED across the board, including all skip_flies_ variants.
- **Beta masquerading as alpha.** On a book losing \$186–195/day, skipping anything "works." Several claimed-ROBUST rules were worse than skipping random days (train MC $p > 0.8$): band_c_0.5-2.5 fly (p 0.998 — the fly stack's #1 member), fly thin-credit floors (p 0.94–0.9999), combined loss-stand-downs (p 0.82–0.90).
- **Fitted families.** skip_event_if_credit_ABOVE and _BELOW both claimed robust on the same ~ 50 event days — the signature of a threshold grid fitted to noise. All 15+ variants refuted.
- **Both-directions VIX rules on ic.** Every ic vol-skip fails (ex-April negative on most); ic day-skips on credit richness fail (half of each test delta is one day). puts_band is hurt by every VIX filter — its profits concentrate on high-vol fat-credit days, consistent with Vilkov's high-VIX-tercile result that binary high-VIX skips discard the best-paid days.
- **Day-after stand-downs on ic/puts.** The day after a killer is a recovery day for ic (avg +\$48, 67% win) and puts_band (+\$100, 67%) — consistent with Kaminski & Lo: stop/halt rules only pay when losses cluster, and ours don't on those books (they do on fly).
- **Pooled circuit breakers.** All fly and pooled ic+fly breakers DEAD at every threshold — fly stop churn trips the trigger 188–424 days and forfeits the settle recoveries that pay for the book.
- **The stacks.** Greedy add/prune keyed on marginal TEST delta = optimizing on the holdout twice; LOYO was measured on membership chosen with all years visible; two unverified components (breaker MID, FOMC bound) sit inside all four stacks. The fly (+180k "recovered", still -20k) and combined (+173.8k, still -36.7k) stacks are bookkeeping on a dead sleeve.

6. Recommended stack per book (with the required caveats)

Do not book the deltas below as validated P&L — membership was test-selected (§5). They are the best current candidate configuration, to be shipped advisory-only and judged forward.

Book	Configuration	Train Δ	Test Δ	Killer days avoided	Book total before $\rightarrow$ after	maxDD
ic	drop call leg if cr_c>2.0/2.5/3.0 + breaker -1000 (stage-2 pending) + total_cr<1.0 floor + FOMC 12:30 exit (stage-2 pending) + loss<-1500-until-VIX-down	+32,53 3	+37,90 4	35	-24.9k $\rightarrow$ +45.5k	32.6k $\rightarrow$ 12.3k
puts_band	total_cr<1.0 floor + breaker -1000 + FOMC 12:30 exit + cr_p<0.5 floor	+5,976	+10,01 0	8	+14.9k $\rightarrow$ +30.9k	8.4k $\rightarrow$ 3.3k
fly	RETIRE (best stack still -20.2k with 903 days zeroed)	—	—	—	-200.5k $\rightarrow$ 0 by not trading	—
combined	do not run a pooled book; run ic + puts_band separately	—	—	—	—	—

Killer-day before/after detail: data/options_sim/backtest_full/killerday/stack_killer_days_20260910.csv (all 193 baseline killer days < -\$1,500, baseline vs stacked); per-rule marginals in stack_rules_20260910.csv; LOYO in stack_loyo_20260910.csv (positive all 20 book-years — with the selection caveat above).

What is adoptable tonight vs pending:

- Adoptable (advisory flags): fly retirement decision on record; puts_band -1000 breaker; dead-tape credit floors as hygiene.
- Pending stage-2 (tomorrow): ic breaker sign, FOMC 12:30 flatten value, call-leg-drop confirmation.
- Live-build gaps found in the executability audit: daemon needs a both-sides pre-fire quote stage, gap-at-fire computation, cross-day cooldown state file, live day-P&L aggregation + flatten-all action; backtest credits are ThetaData marks vs live IB quotes (thresholds transfer approximately).

7. What we cannot fix at day level — and the stage-2 plan

Unfixable at day level: 1. **The 11 ambush days** (§2c): quiet open, calm VIX, no event, kill arrives midday. Only an intraday marked-book breaker, structural sizing, or wing width touches these. Practitioner consensus says sizing is the primary tail control, not stops (Schwartz: "set-and-forget with proper position sizing"; Vilkov: size to expected-shortfall budgets). 2. **The FOMC afternoon.** Statement 13:00 CT, presser 13:30 CT; pre-announcement hours have below-normal vol (Lucca-Moench); ~65% of first moves reverse by the close. A morning entry gate provides zero protection; the intervention must be a 12:30 flatten and/or a post-presser re-entry (OptionAlpha: FOMC days 2.0% vs 1.25% avg move, close at extremes). 3. **Stop-whipsaw losses inside the EM** (2024-12-20, 2026-06-09, 2023-02-14 type): the stop rule, not the market, manufactured the loss. This is a stop-width/credit design question, only testable on intraday marks (double-stop rate ~8.6% and rising per the MEIC live records).

Stage-2 (runs tomorrow, ThetaData terminal free, v2 engine):

- **R5/R6/R7 breaker sim:** 1-min mid marks for our exact historical strikes, trigger at $-2\times/-3\times/-5\times$ morning credit, 1/2/5-min debounce, exits charged at natural + 0.10–0.25/leg slippage, no same-day re-entry; freeze-only vs close-all variants; false-trigger rate by VIX tercile. Decides the ic breaker sign (current bracket OPT +39k/+31k vs PESS -16k/-6k).
- **FOMC 12:30 CT repricing:** actual buyback cost at 12:30 on all 34 FOMC days + optional 13:45 re-entry credit — replaces the accounting bound with a number.
- **Call-leg-drop confirmation:** minute-level v2 rerun with the $cr_c > 2.5$ gate to verify the +7.3k test delta survives real fills.
- **Stop-width study:** per-side stop = total IC credit (MEIC standard) vs current multiples, double-stop frequency measured, on the same marks.

8. Data shopping list (from the research phase)

Item	What it enables	Cost/source
VIX9D daily history	Term-structure inversion gate — best-supported pre-open signal (<u>SSRN 6752518</u> : inversion depth adds 2.4–6.9pp R^2 for forward realized vol)	FREE — <u>Cboe CDN CSV</u>
VIX3M daily history	VIX/VIX3M backwardation regime flag (~7.7% of days; <u>Simon & Campasano</u>)	FREE — <u>Cboe CDN</u>
VVIX daily history	Vol-of-vol divergence early warning (<u>Fed FEDS 2013-54</u>)	FREE — Cboe CDN
SKEW daily history	Low priority; test-to-reject	FREE — <u>Cboe CDN</u>
VIX close 20d-dispersion (already have VIX)	Thrasher "volatility tsunami" complacency trigger (<u>SSRN 2949847</u>) — warns of first-day shocks the coincident family misses	\$0 — computable tonight from data/vix_daily.csv
ES overnight/Globex 1-min bars	Overnight-range gate — the truest pre-open signal (<u>NY Fed SR917</u>); would have caught 04-07-25-class days without VIX	Databento (\$125 credit on file)
ThetaData 1-min SPX option quotes, 2022–2026 strike sets	Stage-2 breaker/FOMC/stop-width sims — the gating purchase ; terminal free after tonight	Existing ThetaData sub
VIX1D history	Per-event richness gate (event-week variance priced ~3.8–4.2× realized vs 1.5–1.6× normal)	FREE — Cboe
Calendar upgrades (no data purchase)	Add: OpEx/quarterly witching, MSCI/S&P rebalance dates, ECB/BOJ, ISM, ECI, Treasury auction slate, US elections, tariff effective-dates. These caused ≥8 of the 33 misses.	\$0 — public schedules

Provenance

- Forensics: 33 per-day investigations, each persisted under scripts/killerday/ with evidence CSVs in data/options_sim/backtest_full/killerday/; web sources cited inline above and in the per-day scripts.
- Sweeps: test_credit.py, test_gap-prior.py, test_vix.py, test_event.py, test_cluster.py, test_breaker.py → family results CSVs (all rules reported including losers).
- Stack: test_stack.py → stack_{steps,final,rules,killer_days,loyo}_20260910.csv.

- Adversarial audit: adversarial_audit.py → adversarial_audit_20260910.csv (174 rule×book recomputes, random-skip MC null, Bonferroni ×395).
- Reproduction audit: all 7 scripts regenerate their CSVs byte-identical; no lookahead found (all inputs prior-close / morning-entry / advance-known calendar).
- Constraints honored tonight: no ThetaData requests, no git commits, no system-state changes.

STAGE-2 UPDATE (2026-09-10, post-workflow): exact intraday circuit-breaker simulation

The day-level breaker bounds above are now superseded by an EXACT event-driven simulation (scripts/killerday/breaker_sim_v2.py): every stop's real timestamp rebuilt from the parity feed, breaker fires when cumulative REALIZED day P&L hits -X, all open legs closed at the real ThetaData touch at the breach minute, entries after breach skipped. Zero missing quotes in the final run. Implementable verbatim in the live daemon (keys on booked losses, not marks).

Results (delta vs baseline, \$ 1-lot, 2022-05-16 -> 2026-09-04; train=22-24 / test=25-26):

book	level	breach days	total	train	test	killer-day delta	verdict
puts_band	-1000	35	+3,342	+942	+2,400	+4,070	ADOPT (advisory) — positive both halves
puts_band	-1500	14	+60	+60	0	+60	neutral
puts_band	-2000/-3000	10/1	0	0	0	0	never binds
ic	-1000	119	-969	-1,596	+627	+5,908	NO — whipsaw pays back the killer savings
ic	-1500/-2000/-3000	67/30/2	-1,473/-639/-104	neg	mixed	neg	NO
combined (incl fly)	-1000	427	-9,683	-12,193	+2,510	+76,352	NO
combined	-1500	199	-720	-3,950	+3,230	+26,138	NO
combined	-2000	106	+4,964	-1,383	+6,347	+11,953	not robust (train negative)
combined	-3000	36	-1,195	-1,220	+25	-1,090	NO

puts_band -1000 breach-day detail (all 35): 9 saves totaling +\$4,070 (2025-03-03 +1,060 tariff-eve, 2026-06-17 +610 FOMC/Warsh, 2024-08-01 +650 ISM, 2022-05-20 +580 OpEx, 2025-02-27 +540, 2026-01-29 +150, 2024-12-18 +60 FOMC, 2025-12-12 +40, 2022-10-14 +380), 2 whipsaw costs (2023-01-03 -478, 2024-03-08 -250), 24 days delta 0 (breach at/near the final exit anyway). Breach times spread 10:33-15:45 CT-shifted-ET — real intraday triggers, not end-of-day artifacts. Full rows: killerday/breaker_v2_days_20260910.csv.

Corrections to the day-level breaker section above: the stage-1 optimistic bounds overstated the ic/combined breaker; exact simulation shows the ic breaker net-negative at every level and any fly-containing aggregate strongly negative at -1000/-1500. The single adoptable breaker is **puts_band at -1000 realized: +\$3.3k/4.3yr** on top of the +\$14.9k base (~+22%), maxDD relief on 9 of the 33 killer days, cost 2 whipsaw days. Advisory-only per the desk rule.